

Learning how to prove Workshop Week 6

Groups



Groups and subgroups

1. Show that there are only two groups of order 4, the cyclic group C_4 and a noncyclic group K_4 , whose Cayley table is as follows:

K_4	1	a	b	c
1	1	a	b	c
a	a	1	c	b
b	b	c	1	a
c	c	b	a	1

Solution: Let $G = \{1, a, b, c\}$ be any group of order 4. We will prove that there are only two ways to fill in the Cayley table of G , provided that G is a group.

First of all notice that the way that 1 multiply is given by the axioms of the group. So, we have that

G	1	a	b	c
1	1	a	b	c
a	a			
b	b			
c	c			

Assume first that $ab = 1$. Then $ac = b$ and so $a^2 = c$. Thus:

G	1	a	b	c
1	1	a	b	c
a	a	c	1	b
b	b			
c	c			

Next $ba = 1$, $bc = a$ and $b^2 = c$.

G	1	a	b	c
1	1	a	b	c
a	a	c	1	b
b	b	1	c	a
c	c			

To finish, $ca = b$, $cb = a$ and $c^2 = 1$. The resulting table is as follows:

G	1	a	b	c
1	1	a	b	c
a	a	c	1	b
b	b	1	c	a
c	c	b	a	1

In this case, we have got that $c = a^2$, $b = ca = a^2 \cdot a = a^3$ and $a^4 = c^2 = 1$, so that $G = \{1, a, a^2, a^3\}$ is cyclic.

If $ac = 1$, then (proceeding like before) we will obtain that $G = \{1, a, a^2, a^3\}$; if $bc = 1$, then $G = \{1, b, b^2, b^3\}$, which is the same group (except for notation). We have then proved that G is cyclic if the product of any two of a , b and c equal 1. Thus, if G is not cyclic is because the product of any of two of a , b , and c must equal the third. In order words:

$$\begin{aligned} ab &= ba = c, \\ ac &= ca = b, \\ bc &= cb = a. \end{aligned}$$

Thus we have that

G	1	a	b	c
1	1	a	b	c
a	a		c	b
b	b	c		a
c	c	b	a	

To finish $a^2 = b^2 = c^2 = 1$, and we get the required Cayley table:

K_4	1	a	b	c
1	1	a	b	c
a	a	1	c	b
b	b	c	1	a
c	c	b	a	1

2. Let H and K be subgroups of a group G .
- (a) Prove that $H \cap K$ is the largest subgroup of G contained in both H and K .
In other words, if L is a subgroup of G such that $L \subseteq H$ and $L \subseteq K$ then $L \subseteq H \cap K$.
- (b) Show that $H \cup K$ is a subgroup of G if and only if $H \subseteq K$ or $K \subseteq H$.

Solution:

(a) In the quiz this week you were asked whether $H \cap K$ is a subgroup of G . The answer is yes, and you should provide a proof in the quizz. Take a look at quiz's memo and let me know whether you need any help with it.

Let L be a subgroup of G such that $L \subseteq H$ and $L \subseteq K$. Then every element $\ell \in L$ is contained in both H and K , which implies that $\ell \in H \cap K$ and so $L \subseteq H \cap K$.

(b) Assume that $H \cup K$ is a subgroup of G . Suppose that $H \not\subseteq K$ and let us show that $K \subseteq H$. Since $H \not\subseteq K$ there exists an element $h \in H$ such that $h \notin K$. On the other hand to prove that $K \subseteq H$ we have to show that $k \in H$ for every $k \in K$. So, take $k \in K$ and consider $kh \in K \cup H$ because $H \cup K$ is a subgroup of G by hypothesis (and k and h are both in $H \cup K$). Notice that $kh \notin K$ because otherwise if

$$kh \in K \Rightarrow k^{-1}(kh) \in K \Rightarrow h \in K, \quad \text{a contradiction.}$$

Thus:

$$kh \in K \cup H, \quad kh \notin K \Rightarrow kh \in H,$$

and therefore $k = (kh)h^{-1} \in H$, as desired.

The converse is clear because if $H \subseteq K$ or $K \subseteq H$ then $K \cup H = K$ or $K \cup H = H$, and so $H \cup K$ is a subgroup of G in any case.

Remark. In general, the union of two subgroups is not a subgroup. For example, consider the additive group $G = \mathbb{Z}$ and take $H = 2\mathbb{Z}$, $K = 3\mathbb{Z}$. Notice that $H \cup K$ is not a subgroup because $2, 3 \in H \cup K$ but $2 + 3 = 5 \notin H \cup K$.